

The empirical recurrence for OEIS A221939, proved by transfer matrix

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Abstract

OEIS A221939 carries an empirical recurrence of order 37 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The entry's condition is local to a bounded window of consecutive lines, so the admissible configurations of such a window are the vertices of a finite digraph and the arrays counted are exactly the walks in it; hence $a(n)$ is a walk count on $S = 5331$ vertices and is C -finite. Whether the conjectured recurrence annihilates it is then settled by a finite exact computation, with the Cayley–Hamilton theorem bounding the work.

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1 The conjecture

OEIS A221939 is “Number of $3 \times n$ arrays of occupancy after each element stays put or moves to some horizontal or antidiagonal neighbor, without move-in move-out straight through or left turns.”. It has offset 1 and begins

1, 91, 1599, 36467, 799632, 17595404, 387146025, 8517616158, ...

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: $a(n) = 6a(n-1) + 286a(n-2) + 1731a(n-3) - 3366a(n-4) - 58755a(n-5) - 73158a(n-6) + 744580a(n-7) + 1606773a(n-8) - 5847271a(n-9) - 13299665a(n-10) + 35585397a(n-11) + 56858689a(n-12) - 168213873a(n-13) - 99886051a(n-14) + 532060381a(n-15) - 127805646a(n-16) - 951724652a(n-17) + 874261131a(n-18) + 729065885a(n-19) - 1511277493a(n-20) + 275583551a(n-21) + 992093720a(n-22) - 708567632a(n-23) - 119679569a(n-24) + 310913349a(n-25) - 92576435a(n-26) - 25624476a(n-27) + 17876348a(n-28) - 1254086a(n-29) - 1085212a(n-30) + 391330a(n-31) - 46391a(n-32) - 25190a(n-33) + 7306a(n-34) + 873a(n-35) - 236a(n-36) - 12a(n-37)$ for $n > 38$

As of the “Last modified” line on the live entry (Oct 03 2025 20:06:56, revision 8) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The count is a walk count

The condition the entry imposes is local: it constrains a bounded window of consecutive lines and nothing beyond. Take as vertices the admissible configurations of one such window and put an edge from u to v whenever v may follow u , which the condition decides by inspection of the two. An admissible array is then exactly a walk, so with M the adjacency matrix on $S = 5331$ vertices and ι, τ the vectors recording which windows may begin and end an array,

$$a(n) = \iota^\top M^{n-1} \tau,$$

